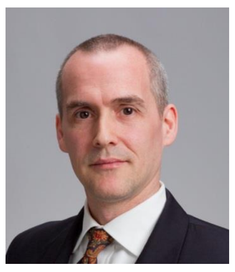
JAIST
JAPAN ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY 1990

Corpus-based analysis of scientific research abstracts



by John Blake

Background

Problems

- Difficulty wording research abstracts
- Patchwork plagiarism trap
- Novice-expert progression
- NNES use of marked phraseology

Importance

- Addresses the problems
- English: *de facto* language of science
- Growth of Information Science in tandem with growth of Internet
- Fills the gap

Gap

- No holistic studies of abstracts
- Few studies of SRA
- No studies of information science RAs
- Answers the calls

Calls

- For domain-specific research by Hyland
- For research on titles by Swales
- For advice on writing SRAs by numerous graduate students

Aim

To help writers avoid accusations of plagiarism and understand the generic expectations in terms of rhetoric organization and lexical realization of scientific research abstracts in the field of information science

Originality

- Interdisciplinary
- Linguistic investigation with pedagogic aims
- Groundbreaking holistic study of SRAs

Significance

- Multiple rich data sets
- Clear problem specification
- Sophisticated analyses

Abstract titles

Pilot study

500 Titles

Structure

Function

Multi-word expressions

1Y Titles

Rhetorical organization

1.1 Structure

1.2 Title in text

1.3 Function

1.4 SFG analysis

Lexical realization

Collocation

1.5 Multi-word expressions

Colligation

1.6 Phrase analysis e.g. countability, modification

Analysis

Title template

Move analysis

Move label set

100 SRA isotextual

2.1 CARS

2.2 IMRaD

2.3 PIMPC

2.4 Aim

Expert analysis

Label set

Manual move identification

500 SRA isotextual

Manual move analysis

Validity

Training set

Automated move identification

1Y SRA isochronic

Automated move analysis

Validity

Analysis

1Y SRA move-tagged

Rhetorical organization

Pilot studies

500 SRA isotextual

Rhetorical organization

1Y SRA move-tagged isotextual, isolexical & isochronic

3.1 Recursivity

3.2 Permutations

3.3 Fronting

3.4 Variation

Analysis

Move template

Lexical realization

1Y SRA move-tagged isotextual, isolexical & isochronic

Pilot studies

Collocation

Colligation

500 SRA isotextual

Lexical realization

4.1 Multi-word expressions

4.2 Clause analysis e.g. type and composition, SVOCA

4.3 Phrase analysis e.g. tense, aspect, voice, valency, transitivity

Analysis

Frequent phrases

Similarity assessment

Frequent phrases

Pilot studies

135 SRA

Similarity assessment

5.1 Verbatim

5.2 Paraphrased

5.3 cf. abstract

Analysis

5.4 Plagiarized

5.5 Cited

5.6 Recyclable

Phrase bank

Abstract Creation Tool

Title template + **Move template** + **Phrase bank** = **Abstract Creation Tool**